

Aspect	Direct Tooling	Indirect Tooling
Process	Tool/mold is created directly from CAD using rapid prototyping (e.g., SLA, SLS)	A master pattern is created first, then used to make the mold (e.g., silicone rubber, epoxy)
Steps Involved	Single-step process (direct fabrication of the mold/tool) ↓	Multi-step process (pattern creation followed by mold making)
Materials Used	Engineering materials directly (e.g., aluminum-filled epoxy, copper polyamide)	Molds are made from softer materials (e.g., silicone rubber, aluminum-filled epoxy)
Time & Cost	Faster and less expensive (since no intermediate steps)	Slower and more expensive due to multiple stages
Suitability	Ideal for short production runs (e.g., 100-400 parts)	Suitable for low-volume production or prototyping
Examples	Direct AIM (ACES Injection Molding), Copper Polyamide Tooling (Direct Rapid Tooling)	Silicone Rubber Tooling, Aluminum-Filled Epoxy Tooling (indirect Rapid Tooling)
Production Volume	Short runs (100-400 parts)	Limited production (up to 50 cycles with materials like silicone)
Application	Quick iterations, prototyping, reduced time to production	More suited for complex geometries or delicate designs
Complexity Handling	Generally simpler shapes and designs	Better for complex or intricate shapes
Advantages	Faster turnaround, minimal intermediate steps	More flexibility with design intricacies, supports complex shapes
Example Materials	Aluminum-filled epoxy, copper polyamide (Direct Rapid Tooling)	Silicone rubber, aluminum-filled epoxy, spray metal tooling (indirect Rapid Tooling)

Explain part building errors in SLS process and how to compensate for it.

i) due to shrinkage
when the powdered material is heated it melts and fuses. and as it cools it shrinks. This leads to uneven shrinkage and hence the final part may not match the ~~inted~~ intended size and shape.

ii) areas with high temp. can shrink more than those with lower temp. and parts with thick walls or sections can increase the shrinkage.

— To compensate

i) Material coefficient is calculate using test part and scaling factor is applied to each direction in STL file

ii) scaling and offsetting are applied to part dimensions according to foll. relation.

$$\text{New dimension} = a \left(\text{desired dimension} \right) + b$$

/ scaling factor
+ b — offset

3

- **Improves Surface Finish and Resolution:** Proper orientation reduces the stair-step effect, leading to smoother surfaces and finer details.
- **Reduces Build Time:** Optimal orientation minimizes the number of layers and supports, speeding up the printing process.
- **Minimizes Material Usage:** Correct orientation reduces the need for support structures, saving material and simplifying post-processing.
- **Prevents Trapped Volumes:** Proper orientation eliminates trapped liquid in liquid-based printing processes, ensuring defect-free prints.
- **Enhances Mechanical Strength:** Aligning the orientation with load-bearing directions improves the part's strength and durability.
- **Reduces the Stair-Step Effect:** Thinner layers are used in slanted surfaces, minimizing visible layer lines and improving accuracy.
- **Supports Protruding Features:** Correct orientation minimizes the need for support structures on cantilevers or overhangs, improving structural integrity.

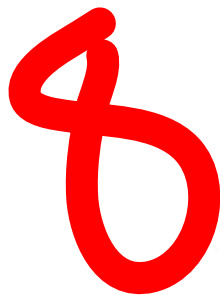
4

Criteria	Silicone Rubber Tooling	Aluminum Epoxy Tooling
Material	Silicone Rubber	Aluminum-filled Epoxy Resin
Fabrication Process	Rubber is blended, vacuumed, and poured into a box around the pattern. The mold is then cut into two halves after curing.	Epoxy resin is poured around a master pattern in a wooden frame, with metal inserts and cooling lines sometimes added. The second half is cast after the first half hardens.
Pattern Creation	The RP (Rapid Prototyping) pattern is encapsulated in silicone.	The RP pattern is embedded in the epoxy mold.
Curing Time	12-24 hours for silicone rubber to cure.	Epoxy resin hardens before the second half is cast.
Number of Uses	Can be used up to 50 times.	Can withstand more cycles due to the strength of the epoxy, especially when reinforced with metal inserts.
Material Injection	Not injected due to the softness of the material; parts are poured instead.	Suitable for injection molding due to the rigid nature of the epoxy.
Strength	Soft and flexible, unsuitable for high-pressure processes.	High strength, can withstand higher pressures in injection molding.
Surface Finish	Suitable for smooth finishes and fine details.	Suitable for larger parts and complex geometries but may need extra support.
Cooling	No cooling lines included.	Water cooling lines are often included during fabrication.

1. **Pattern Creation:** Sand casting uses a model to create an impression in two halves of a sand mold, called the cope and drag.
2. **Material Used:** Sand mixed with bonding agents forms the mold, which is contained in a flask to hold its shape.
3. **Molten Metal Pouring:** Molten metal is poured into the sand impression and allowed to cool before being pulled from the mold.
4. **Use of RP Patterns:** Rapid prototyped (RP) models can be used as patterns for sand casting, sometimes lasting for several castings before distortion occurs.
5. **Dimensional Accuracy:** While sand casting may result in slightly lower dimensional accuracy than investment casting, it is useful for producing test articles or hardware with machining allowances.
6. **Complexity Limitations:** Sand casting has limitations in handling complex geometries, sharp edges, or intricate internal structures, as these can disrupt the mold.
7. **Versatility:** This process is suitable for producing small to large quantities of parts, but the pattern design must allow for easy removal from the mold without damaging the surface.

3D ~~add~~ lctool

- This process is based on metal sintering process.
- developed 25 years ago
- presently owned & lg licensed by 3D system
- used for creating too moderate to high volume tool
- Any type of RP generated patterns can be used.
- It offers good accuracy & finish from long lasting steel tool.
- The process starts with RP patterns from which rubber mould is created.
- /This rubber ~~mould~~ is then filled with a mixture of steel powder & polymer binder
Then it hardens to green state
- Then the green mould is heated to remove the binder ,
Then copper infiltration is done when ~~the~~ the copper fills the spaces between the steel particles resulting with composition (70% steel & 30% copper)
- The tool may undergo final machining & polishing where ejector pins, cooling lines and other accessories may be added.



3) What is tessellation? What are the errors in tessellation

- It is a process of approximating a 3D CAD models surface with a series of connected triangles
- It is commonly used in rapid prototyping & 3D printing where the original design must be converted to STL format for printing.

Errors in tessellation

- Surface is not perfectly smooth.
 - When curved surface is broken into triangles the shape is not the same as original
- Adding more triangles may make the shape smoother but it can make the file larger
- If there are fewer triangles than, the shape may look rough and reduce accuracy.

9

Direct AIM method (Acres injection moulding)

It is an imp rapid tooling process for building prototype parts using thermoplastics in very short time and inexpensively.

Steps

- i) - creating a mould using stereolithography
- the mold ^{halves} is ~~designed~~ made with hollow cavities to form necessary part shapes.
- ii) These hollow cavities are filled with materials like ceramics, aluminium filled epoxy or low melting metals.
- iii) This backfilling process provides strengthen the moulds and helps with heat transfer for cooling process.
- iv) Then the mold halves are aligned and surface finishing is done.
- v) Then various parts can be made by directly injecting the thermoplastic in ACES mold and cavity halves using standard injection molding machine.

8) Copper polyamide tooling

- It is a rapid tooling method that combines copper and plastic to create durable, heat resistant mold ~~with~~ for short production run.
- Copper polyamide tooling uses composite material made up of copper and polyamid (plastic)
- This composite material is thermally conductive and ~~high~~ heat resistant.

Steps

i) Creating tool inserts.

- Inserts are built directly using SLs machine with 75 micron layer thickness
- Cooling lines, ejector pins guides, gates and runners are designed and printed directly into tool inserts.

ii)

- After printing, the inserts are finished by sealing with epoxy and polishing with sand paper.
- The finished inserts are packed with metal alloys to further strengthen the ~~the~~ molds

8) Explain the data preparation errors in rapid prototyping.

① i) Tessellation errors (Ans in Q*)

② ii) Slicing errors.

Here the 3D model is sliced into thin horizontal layers for printing

- Hence these RP processes causes stair stepping ~~effect~~ problem especially on slanted or curved ~~areas~~ surfaces
- Thicker layer prints faster but makes the stair steps effect more obvious and makes the surface rough.
- Thinner layer gives smoother results but takes more time to print.

③ iii) Part building errors

a) ~~●~~ Curing errors:

- Occurs when over curing of material takes places
- that is when it is exposed to more heat or laser than required
- this causes extra material to harden
- this causes dimensional inaccuracies with the part becoming slightly larger or deformed.

12

b) Control errors :

- errors caused when the machine does not precisely follow layer thickness ~~and~~ or position
- If the thickness varies from layer to layer the part will not have intended dimensions
- If the machines laser or nozzle moves slightly off-position, the parts border can shift, leading to misaligned layers.

8) Orientation Constraints of SLG process
Depends on 2 factors

i) the process accuracy in xy plane and z direction.

ii) Material used (material include polycarbonates, polyamides, elastomers, metal, sands) can behave differently so orientation depends on how much they shrink.

iii) Maximizing strength.

To make the part stronger, they have to be build in x-y plane. This gives more flexibility and reduces chance of breaking

iv) To have best feature definⁿ the critical features should face upwards

v) The parts cross sectional areas should change gradually to avoid shrinkage problems hence the parts are angled by $10-15^\circ$

vi) Parts should be printed in the same orientation that it was cooled to avoid warping (bending after printing)

* NOT SURE ,
14).

Electron beam melting

Advantages

- i) parts can be built with support structures
- ii) EBM operates in vacuum, ~~the~~ reducing contamination
- iii) EBM can create complex structure
- iv) EBM can work with wide range of metal powders

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Disadvantages

- i) EBM can be slower compared to other manufacturing processes
- ii) It is costly

14

15

Solid ground Curing

- This process has 2 main cycles

- i) photomask cycle
- ii) fabrication cycle.

This process takes 2 mins to complete all layers operation to make a single layer.

Steps in SGC process

- i) Lead films are sliced
- ii) Layer of photo sensitive polymer is sprayed onto the platform.
- iii) The photomask is prepared for each layer using electrophotography / Xerography
- iv) The mask plate is placed on top of the platform and object under construction.
- v) UV light is exposed onto the layer, which solidifies completely.
- vi) The residual polymer is vacuumed and replaced by wax.
- vii) The wax is hardened using the wax cooling plate.
- viii) A milling head is used onto the photopolymer to make sure that there is uniform thickness in order to prepare the next layer.
- ix) The mask plate is raised and reused after every layer.
- x) The process is repeated until the object is fully built.